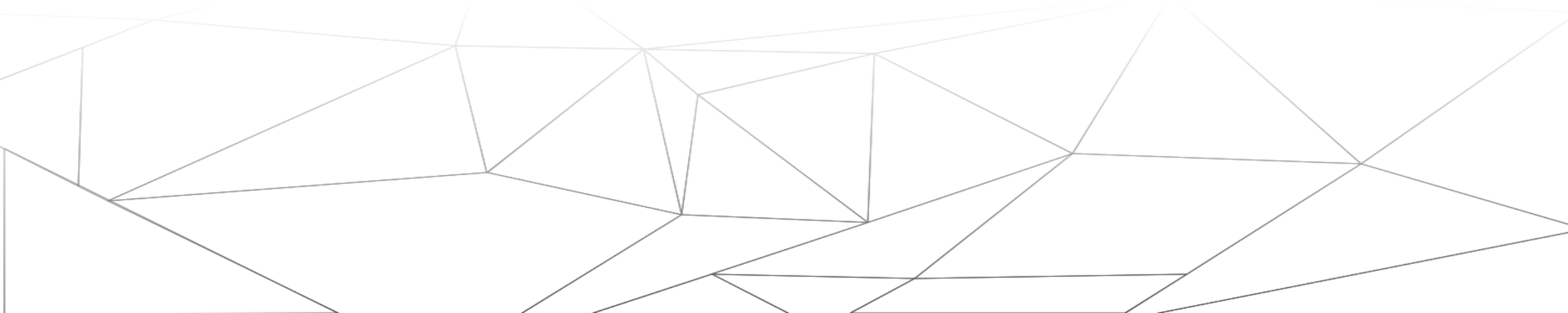


Dark Data

Svea Eckert – Andreas Dewes



Who we are

Svea Eckert

Journalist (NDR/ ARD)



@sveckert

Andreas Dewes

Data Scientist (DCSO & 7scientists)



@japh44

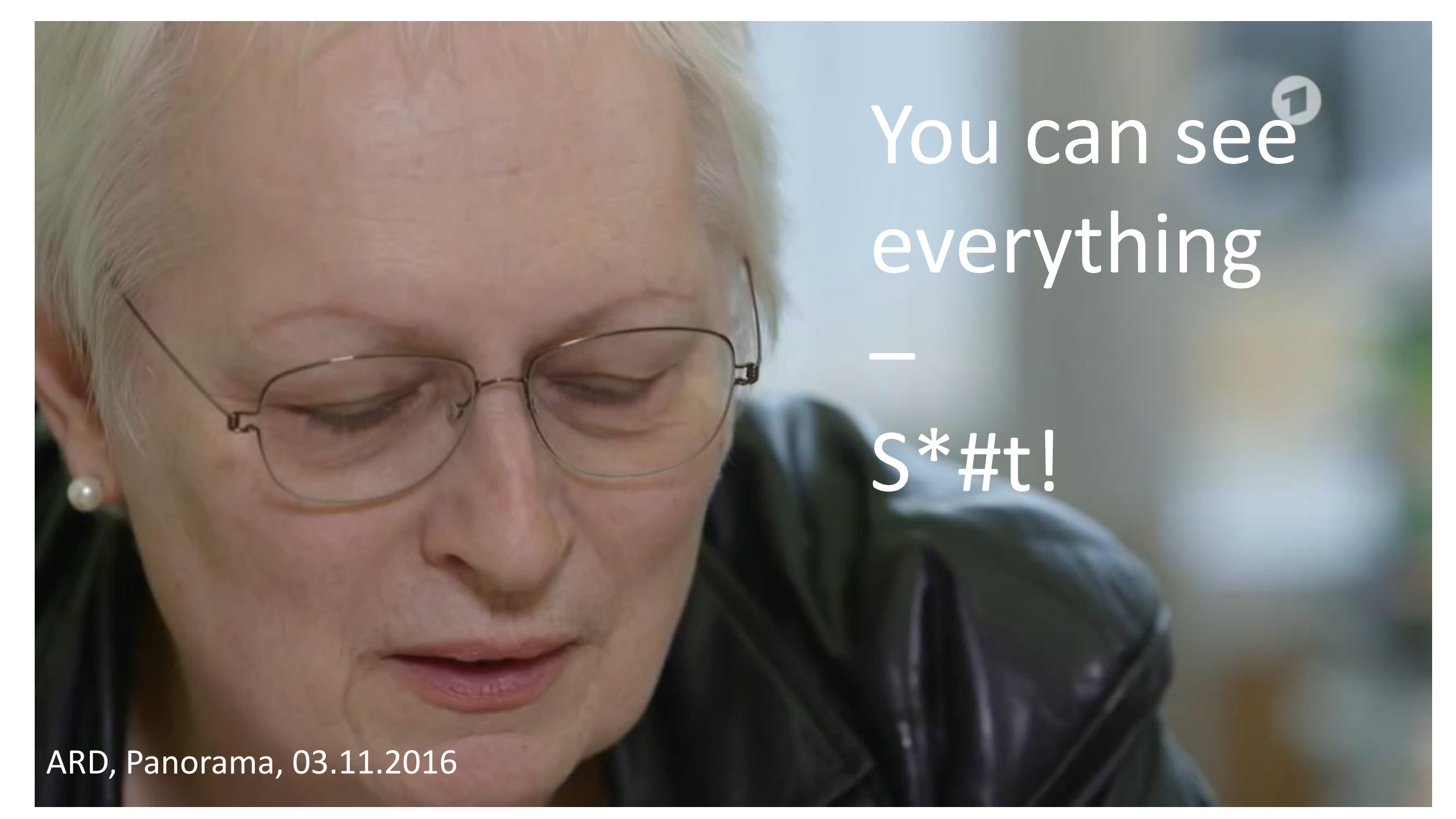
Why we are here

*“US Senate voted to eliminate **broadband privacy rules** that would have required ISPs to get consumers' explicit consent before selling or sharing **Web browsing** data (...)”*

3/23/2017

<https://arstechnica.com>

What does that mean



You can see¹
everything

—

S*#t!

01.08.16 06:11:48 <https://banking.postbank.de/rai/crypt/kCvFlnxhWfHAIJhNSUZn9QVwsNnwtrB9mIJHL6OqFglQ9L>

01.08.16 06:11:59 <https://banking.postbank.de/CvFlnxhWfHAIJhNSUZn9QVwsNnwtrB9mIJHL6OqFglQ9L>

01.08.16 06:12:06 <https://banking.postbank.de/rai/crypt/kCvFlnxhWfHAIJhNSUZn9QVwsNnwtrB9mIJHL6OqFglQ9L>

01.08.16 06:12:45 <https://banking.postbank.de/rai/logout>

01.08.16 06:13:10 <https://www.schwaebisch-hall.de/>

01.08.16 06:13:16 <https://www.schwaebisch-hall.de/mein-konto.html#contracts>

01.08.16 06:13:24 <http://www.welt.de/>

01.08.16 06:13:43 <http://www.welt.de/>

01.08.16 06:23:55 <http://www.sueddeutsche.de/>

01.08.16 06:24:42 <https://outlook.live.com/owa/?path=/mail/inbox/rp>

01.08.16 06:25:47 <http://www.spiegel.de/auto/fahrkultur/gm-futurliner-der-science-fiction-bus-von-general-motors-a.html>

01.08.16 06:26:27 <http://www.spiegel.de/>

01.08.16 06:27:38 <http://www.spiegel.de/politik/ausland/news-des-tages-sigmar-gabriel-horst-seehofer-erdogan-demo-a.html>

01.08.16 06:27:47 <http://m.welt.de/wirtschaft/article157421718/Der-gefaehrliche-Geiz-im-Containergeschaeft.html>

01.08.16 06:27:53 <http://m.welt.de/wirtschaft/article157421718/Der-gefaehrliche-Geiz-im-Containergeschaeft.html>

02.08.16 16:47:34 <http://www.welt.de/wirtschaft/article157421718/Der-gefaehrliche-Geiz-im-Containergeschaeft.html>

02.08.16 16:47:46 <http://www.welt.de/wirtschaft/article157421718/Der-gefaehrliche-Geiz-im-Containergeschaeft.html?config=print>

02.08.16 16:47:50 <http://www.welt.de/wirtschaft/article157421718/Der-gefaehrliche-Geiz-im-Containergeschaeft.html>

02.08.16 16:48:32 <https://www.freitag.de/autoren/felix-werdermann/das-autonome-auto>

02.08.16 16:53:36 <https://banking.postbank.de/rai/login>

02.08.16 17:00:09 <https://banking.postbank.de/rai/crypt/25KzWw0hDqwcUJC5ZH96GpdbmQV6DusTifPKNOk4CUQ>

02.08.16 17:00:22 <https://banking.postbank.de/rai/crypt/6BHP>

02.08.16 17:00:28 <https://banking.postbank.de/rai/logout>

02.08.16 17:00:45 <https://www.schwaebisch-hall.de/>

02.08.16 17:00:51 <https://www.schwaebisch-hall.de/mein-konto.html#contracts>

02.08.16 17:00:57 <https://www.schwaebisch-hall.de/hinweise/logout.html>

02.08.16 17:01:41 <http://www.spiegel.de/>

02.08.16 17:01:50 <http://www.spiegel.de/politik/deutschland/petra-hinz-warum-gibt-spd-abgeordnete-nach-dem-betrug-ihr-mandat-nicht-zurueck-a.html>

02.08.16 17:01:56 <http://www.spiegel.de/>

02.08.16 17:05:51 <http://www.faz.net/>

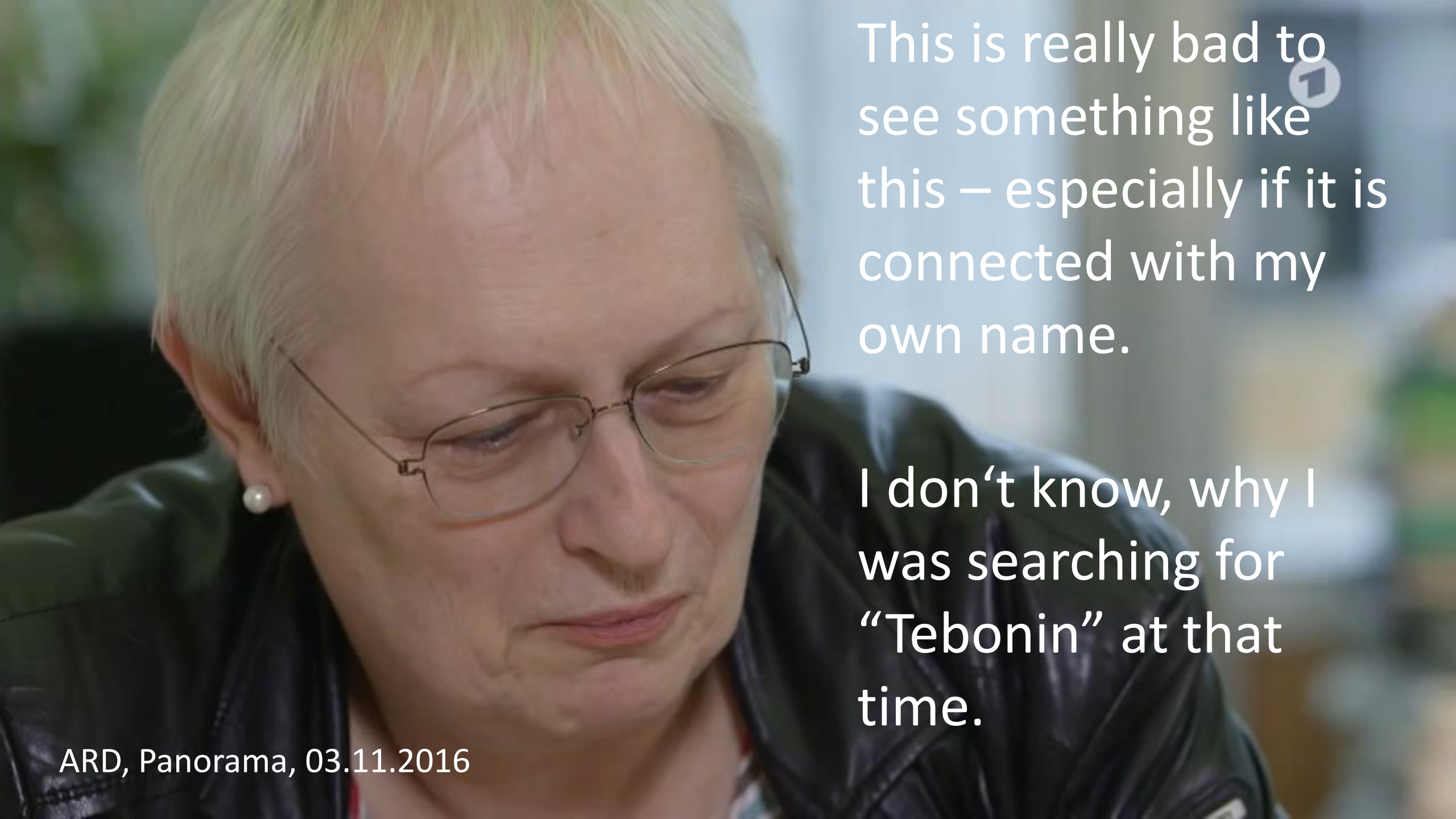
02.08.16 17:06:32 <http://www.faz.net/aktuell/wirtschaft/vw-abgasskandal/abgas-skandal-gruene-kritisieren-csu-setzt-euch-fuer-deutsche-vw-kunden-ein-14368638.html>

02.08.16 17:06:45 <http://www.faz.net/>

02.08.16 17:07:44 <http://www.faz.net/aktuell/wirtschaft/vw-abgasskandal/volkswagens-abgas-skandal-hannover-an-der-leine-14368776.html>

24.08.16 11:47:44 http://www.sueddeutsche.de/wirtschaft/automatuelle-sieg-fuer-die-zuenerer-1.5102072
24.08.16 11:48:03 http://www.sueddeutsche.de/
24.08.16 11:50:07 https://www.elsteronline.de/eportal/Oeffentlich.tax
24.08.16 11:51:24 https://www.elsteronline.de/eportal/KonfigurationsAssistent1JS.tax?requestaction=KonfigurationsAssistent1JS.tax&action=/eop/auth/LoginSoft
24.08.16 11:51:29 https://www.elsteronline.de/eportal/KonfigurationsAssistent1JS.tax?ElsterRequestKeys.KONFIGURATIONS_ASSISTENT_WORKFLOW=SQ&web
24.08.16 11:51:30 https://www.elsteronline.de/eportal/SwitchModule.tax?switchpage=/Postfach.tax&switchprefix=/all/postfach
24.08.16 11:52:28 https://www.elsteronline.de/eportal/eop/formular/Formulare.tax?menuKey=formulare&org.apache.struts.taglib.html.TOKEN
24.08.16 11:53:06 https://www.elsteronline.de/eportal/eop/formular/Formulare.tax?menuKey=estformulare&org.apache.struts.taglib.html.TOKEN
24.08.16 11:53:17 https://www.elsteronline.de/eportal/eop/formular/est/FormularEst.tax
24.08.16 11:53:22 https://www.elsteronline.de/eportal/eop/formular/est/y2015/FormulardatenFruehereAbgabenPrepare.tax
24.08.16 11:54:35 https://www.elsteronline.de/eportal/eop/formular/est/y2015/FormulardatenFruehereAbgaben.tax
24.08.16 11:54:49 https://www.elsteronline.de/eportal/eop/formular/est/y2015/FormulardatenFruehereAbgabeLaden.tax?aufgabeld=
24.08.16 11:54:50 https://www.elsteronline.de/eportal/eop/formular/est/y2015/FormularEst.tax
24.08.16 11:55:26 https://www.elsteronline.de/eportal/eop/formular/est/y2015/FormularEst.tax#modal-speichernUndVerlassen
24.08.16 11:55:54 https://www.elsteronline.de/eportal/SwitchModule.tax?switchpage=/Postfach.tax&switchprefix=/all/postfach
24.08.16 11:56:01 https://www.elsteronline.de/eportal/eop/formular/Formulare.tax?menuKey=formulare&org.apache.struts.taglib.html.TOKEN=
24.08.16 11:56:07 https://www.elsteronline.de/eportal/eop/formular/Formulare.tax?menuKey=euerformulare&org.apache.struts.taglib.html.TOKEN=
24.08.16 11:56:10 https://www.elsteronline.de/eportal/eop/formular/euer/FormularEuer.tax
24.08.16 11:56:20 https://www.elsteronline.de/eportal/eop/formular/euer/y2015/FormulardatenFruehereAbgabenPrepare.tax
24.08.16 11:57:05 https://www.elsteronline.de/eportal/eop/formular/euer/y2015/FormulardatenFruehereAbgaben.tax
24.08.16 11:57:14 https://www.elsteronline.de/eportal/eop/formular/euer/y2015/FormulardatenFruehereAbgabeLaden.tax?
24.08.16 11:57:15 https://www.elsteronline.de/eportal/eop/formular/euer/y2015/FormularEuer.tax
24.08.16 11:57:25 https://www.elsteronline.de/eportal/eop/formular/euer/y2015/FormularEuer.tax#modal-speichernUndVerlassen
24.08.16 11:57:37 https://www.elsteronline.de/eportal/SwitchModule.tax?switchpage=/Postfach.tax&switchprefix=/all/postfach
24.08.16 11:57:40 http://www.taz.de/pt/.nf/home
24.08.16 11:57:57 https://www.taz.de/
24.08.16 12:05:39 https://www.taz.de/Politik/lp4615/
24.08.16 12:06:30 https://www.elsteronline.de/eportal/Logout.tax
24.08.16 12:06:39 http://www.welt.de/
24.08.16 12:26:50 http://www.abendblatt.de/
24.08.16 12:27:00 http://www.abendblatt.de/
24.08.16 12:27:16 http://www.bild.de/wa/ll/bild-de/unangemeldet-42925516.bild.html

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18.08.16	18:39:05	http://www.sueddeutsche.de/
18.08.16	18:39:45	http://www.sueddeutsche.de/wirtschaft/autoindustrie-vw-will-mit-voller-haerte-gegen-streikende-zulieferer-vorgehen-1.3126280
18.08.16	18:39:53	http://www.sueddeutsche.de/wirtschaft/autoindustrie-vw-will-mit-voller-haerte-gegen-streikende-zulieferer-vorgehen-1.3126280-2
18.08.16	18:41:37	http://www.handelsblatt.com/
18.08.16	18:43:22	http://www.handelsblatt.com/unternehmen/industrie/volkswagen-werke-vw-schickt-ueber-20-000-beschaefigte-in-kurzarbeit/14026328.html
18.08.16	18:43:24	http://teslamag.de/
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18.08.16	18:45:17	http://teslamag.de/
18.08.16	18:46:07	http://teslamag.de/news/model-3-umgeklappte-rueckbank-wird-eine-ebene-ladeflaeche-bieten-9332
18.08.16	18:46:14	http://teslamag.de/news/tesla-wird-die-eigenen-stores-radikal-umgestalten-sagt-verkaufsleiter-9329
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18.08.16	18:47:18	https://www.tesla.com/de_DE/models/design
18.08.16	18:47:27	} http://www.sanego.de/Medikamente/Tebonin/
18.08.16	18:48:43	
18.08.16	18:48:43	
18.08.16	18:49:28	https://duckduckgo.com/?q=tebonin+wikipedia&atb=v19&ia=web
18.08.16	18:49:36	https://de.wikipedia.org/wiki/Bilobalid
18.08.16	18:49:53	https://de.wikipedia.org/wiki/Dr._Willmar_Schwabe
18.08.16	18:52:38	http://www.weser-kurier.de/
19.08.16	07:18:41	http://www.weser-kurier.de/region/osterholzer-kreisblatt.html
19.08.16	07:37:29	https://duckduckgo.com/?q=b74-g10&atb=v19&ia=web
19.08.16	07:38:23	https://r.duckduckgo.com/l/?kh=-1&uddg=http%3A%2F%2Fwww.bvwp-projekte.de%2Fstrasse%2FB74-G10-NI%2FB74-G10-NI.html
19.08.16	07:38:27	http://bvwp-projekte.de/
19.08.16	07:38:28	https://duckduckgo.com/?q=b74-g10&atb=v19&ia=web
19.08.16	07:38:31	https://duckduckgo.com/?q=b74-g10+niedersachsen&atb=v19&ia=web
19.08.16	07:38:54	https://r.duckduckgo.com/l/?kh=-1&uddg=http%3A%2F%2Fwww.bvwp-projekte.de%2Fstrasse%2Fdownload_plaene%2FNI%2FB74-G10-NI%2FLPL_2
19.08.16	07:39:00	http://bvwp-projekte.de/
19.08.16	07:39:01	https://duckduckgo.com/?q=b74-g10+niedersachsen&atb=v19&ia=web
19.08.16	07:39:03	https://duckduckgo.com/?q=landesvermessungsamt+niedersachsen&atb=v19&ia=web
19.08.16	07:40:07	https://r.duckduckgo.com/l/?kh=-1&uddg=http%3A%2F%2Fwww.lgn-niedersachsen.de%2Fstartseite%2F



This is really bad to see something like this – especially if it is connected with my own name.

I don't know, why I was searching for "Tebonin" at that time.



Employee of Helge Braun, CDU – Assistant Secretary of the German Chancellor

How we did it – the “hacking” part

Social engineering

The background of the slide features a complex, abstract geometric pattern. It consists of numerous thin, dark gray lines that intersect to form a variety of triangles and polygons of different sizes. These shapes are layered and overlap, creating a sense of depth and a modern, architectural feel. The overall composition is minimalist and clean, with the text 'Social engineering' placed in the lower-left quadrant.



Search for people, jobs, companies, and more...



Advanced



2



Home

Profile

My Network

Learning

Jobs

Interests

Business Services

Try Premium for free

Edit Background



Anna Rosenberg

Senior Consultant bei Meez Technology

Tel Aviv - Jaffa, Tel Aviv District, Israel | Marketing and Advertising

Current Meez Technology

Education Universität Hamburg

View profile as



[Update your public profile settings](#)

Contact Info

No, do not publish an update to my network about my profile changes.



No

104
connections

Add a section to your profile – be discovered for your next career step.

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DATA DRIVEN CONSULTING

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About

WHAT WE DO

PRODUCTS

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CAREERS

About

WHAT WE DO
PRODUCTS

CAREER AT MEEZ

We are constantly looking for creative minds, analysts and dedicated doers for our team. We can

Typekit by Adobe

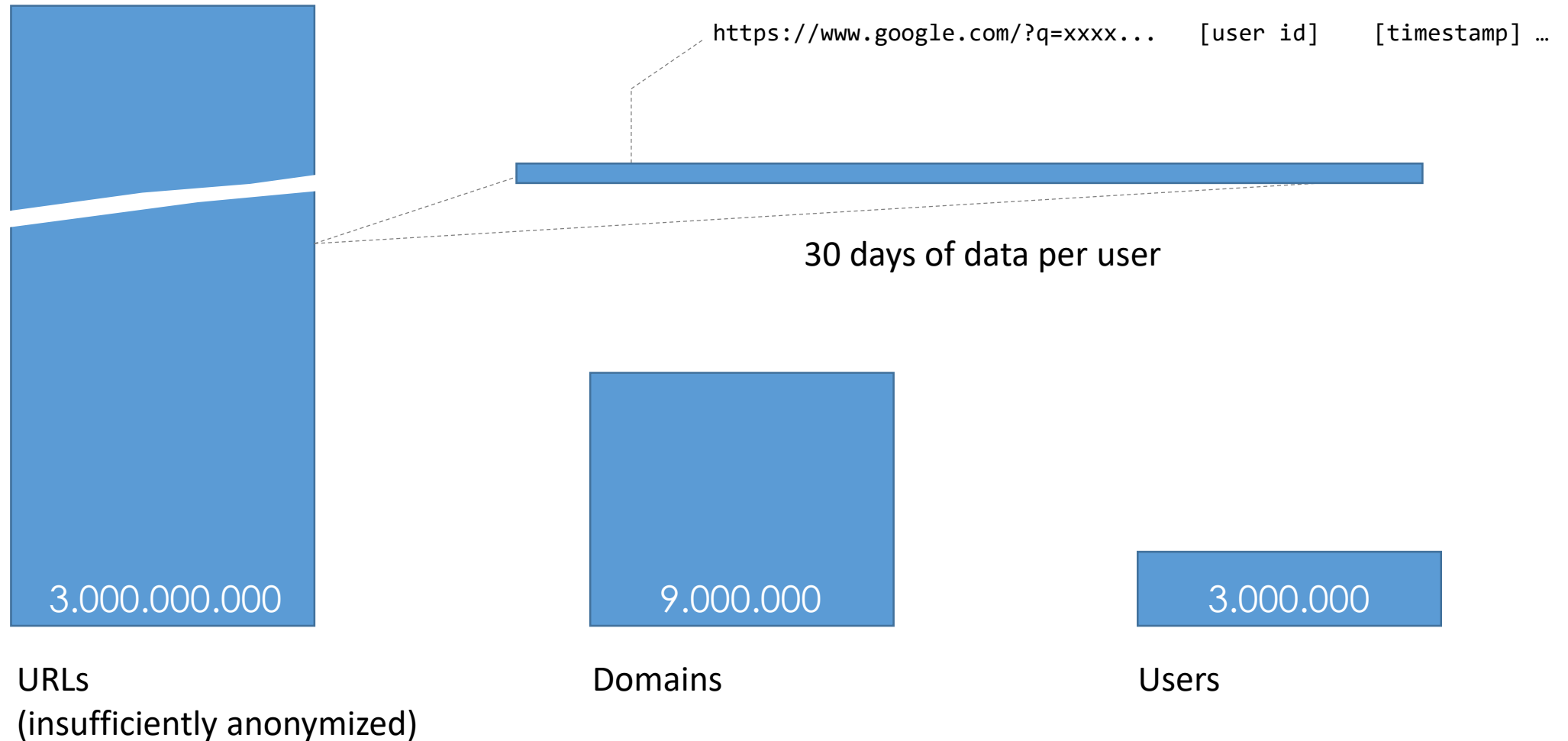
CONTACT US

If you are just as excited about the potential of technology and creativity, then we'd love to hear from you. We are a small, international team, based in beautiful and prospering Tel Aviv.

Mail: mail@meez.tech

What did we get?

Our data set



Statistical Deanononymization

https://www.cs.cornell.edu/~shmat/shmat_oak08netflix.pdf

Robust De-anonymization of Large Datasets (How to Break Anonymity of the Netflix Prize Dataset)

Arvind Narayanan and Vitaly Shmatikov

The University of Texas at Austin

February 5, 2008

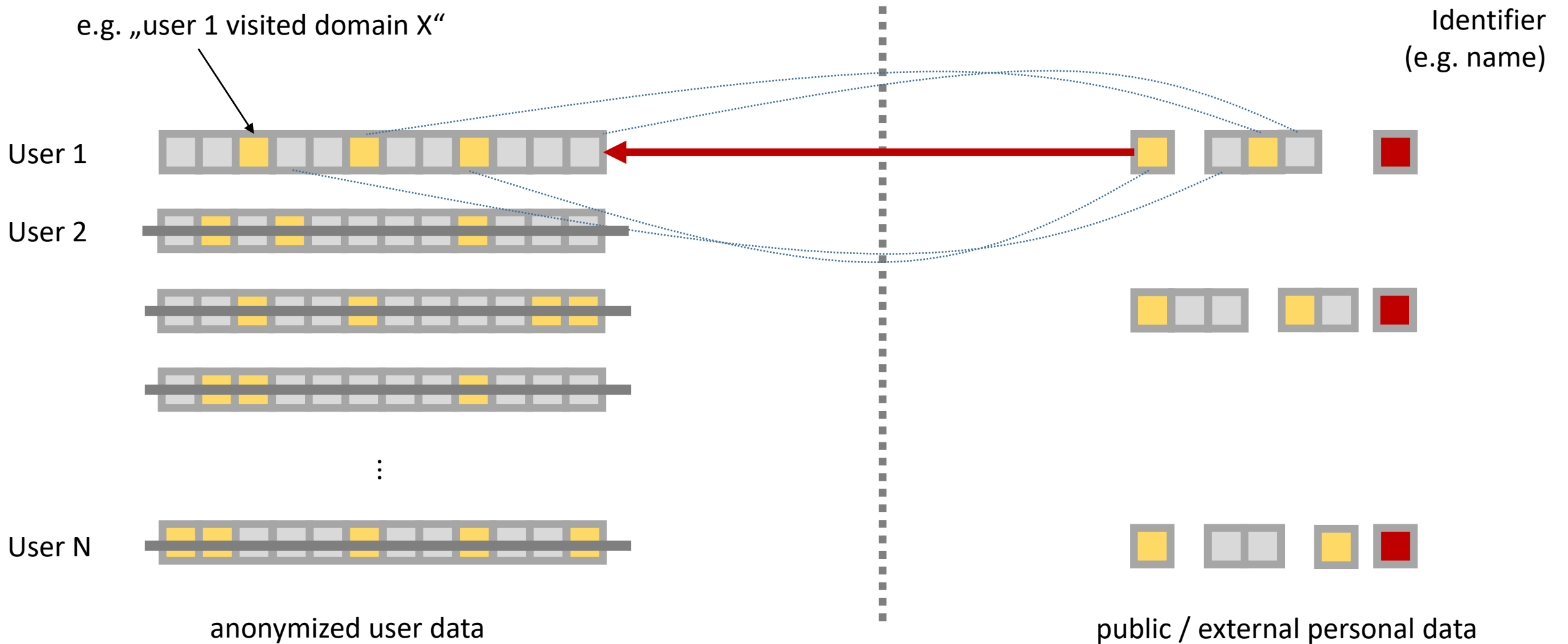
Abstract

We present a new class of statistical de-anonymization attacks against high-dimensional micro-data, such as individual preferences, recommendations, transaction records and so on. Our techniques are robust to perturbation in the data and tolerate some mistakes in the adversary's background knowledge.

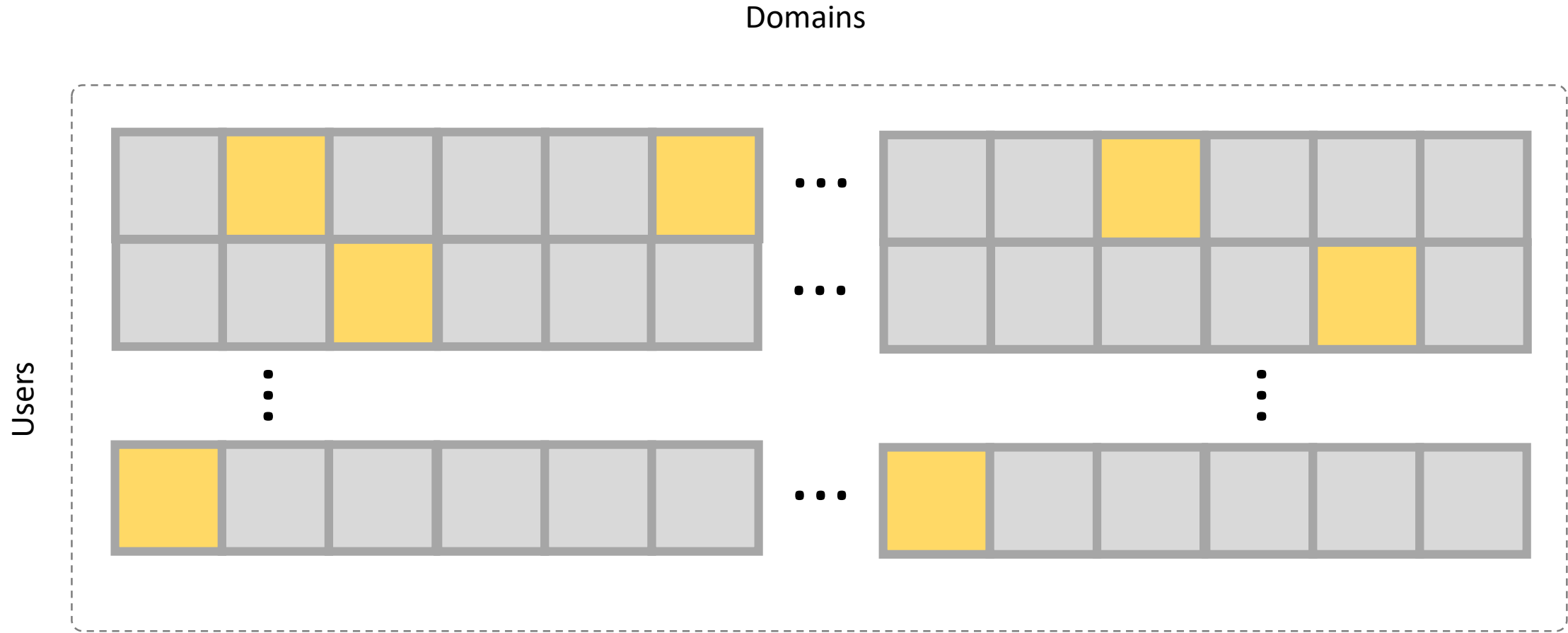
We apply our de-anonymization methodology to the Netflix Prize dataset, which contains anonymous movie ratings of 500,000 subscribers of Netflix, the world's largest online movie rental service. We demonstrate that an adversary who knows only a little bit about an individual subscriber can easily identify this subscriber's record in the dataset. Using the Internet Movie Database as the source of background knowledge, we successfully identified the Netflix records of known users, uncovering their apparent political preferences and other potentially sensitive information.

[cs.CR] 22 Nov 2007

How does it work?



Let's try this!



=> sparsely populated matrix with **9.000.000 x 1.000.000** entries

Our Algorithm

- Generate user/domain matrix M
- Generate vector v with information about visited domains
- Multiply $M \cdot v$
- Look for best match

$$M = (...)$$

$$w = M \cdot v$$

$$i = \operatorname{argmax} (w)$$

How well does this work?

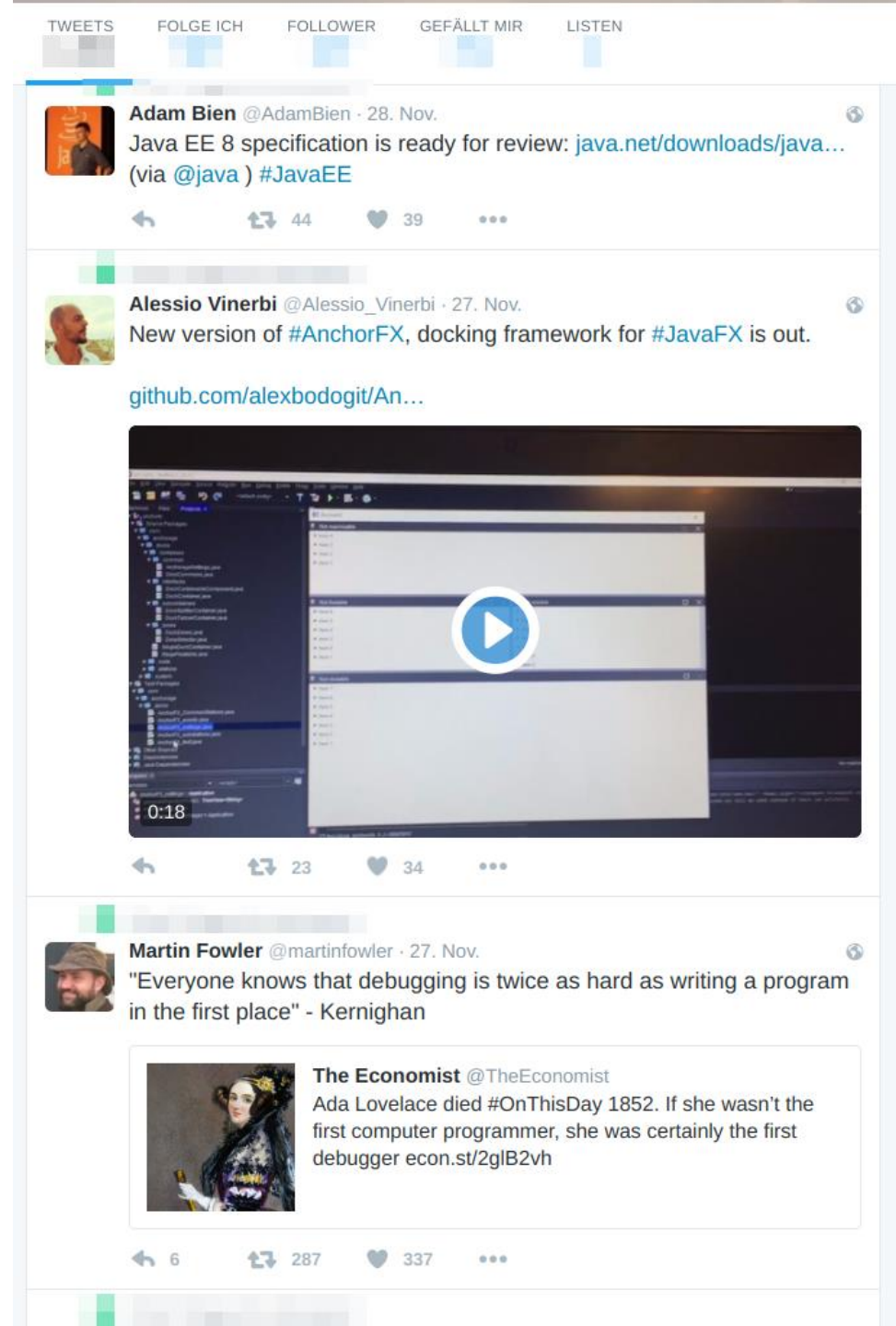


But how can public information be extracted?

Two examples

Twitter

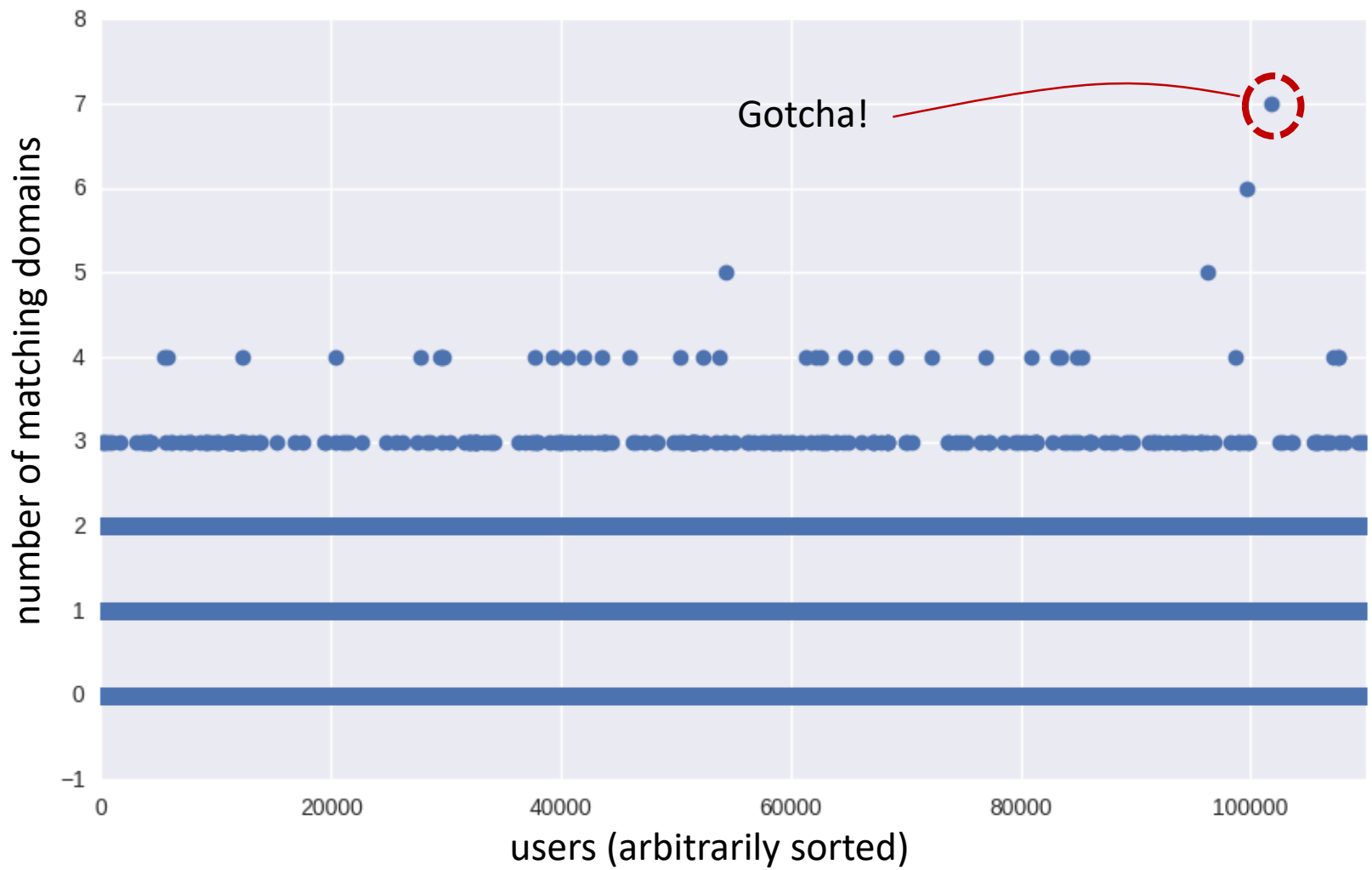
- We use the Twitter API to download tweets from the relevant time period (one month)
- We extract URLs from the tweets and generate the associated domain by following the links
- We feed the domain information into our algorithm



Visited Websites

github.com (2.584.681)
www.change.org (124.152)
fxexperience.com (394)
community.oracle.com (5161)
paper.li (2689)
javarevisited.blogspot.de (525)
www.adam-bien.com (365)
rterp.wordpress.com (129)

Examples

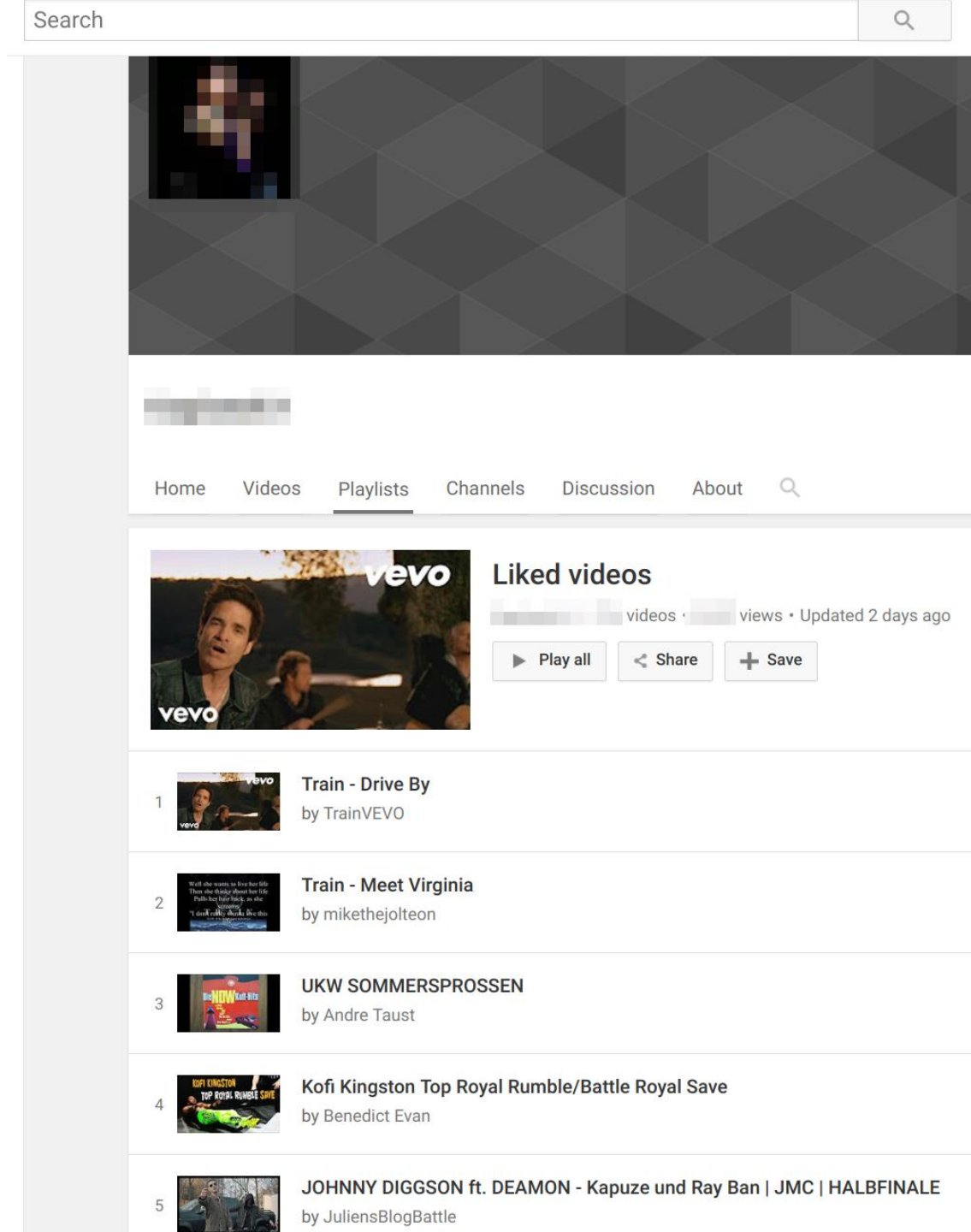


Seemingly harmless identifiers can betray you

[https://www.youtube.com/watch?v=DLzxrzFCy0s](https://www.youtube.com/watch?v=<u>DLzxrzFCy0s</u>)

Youtube

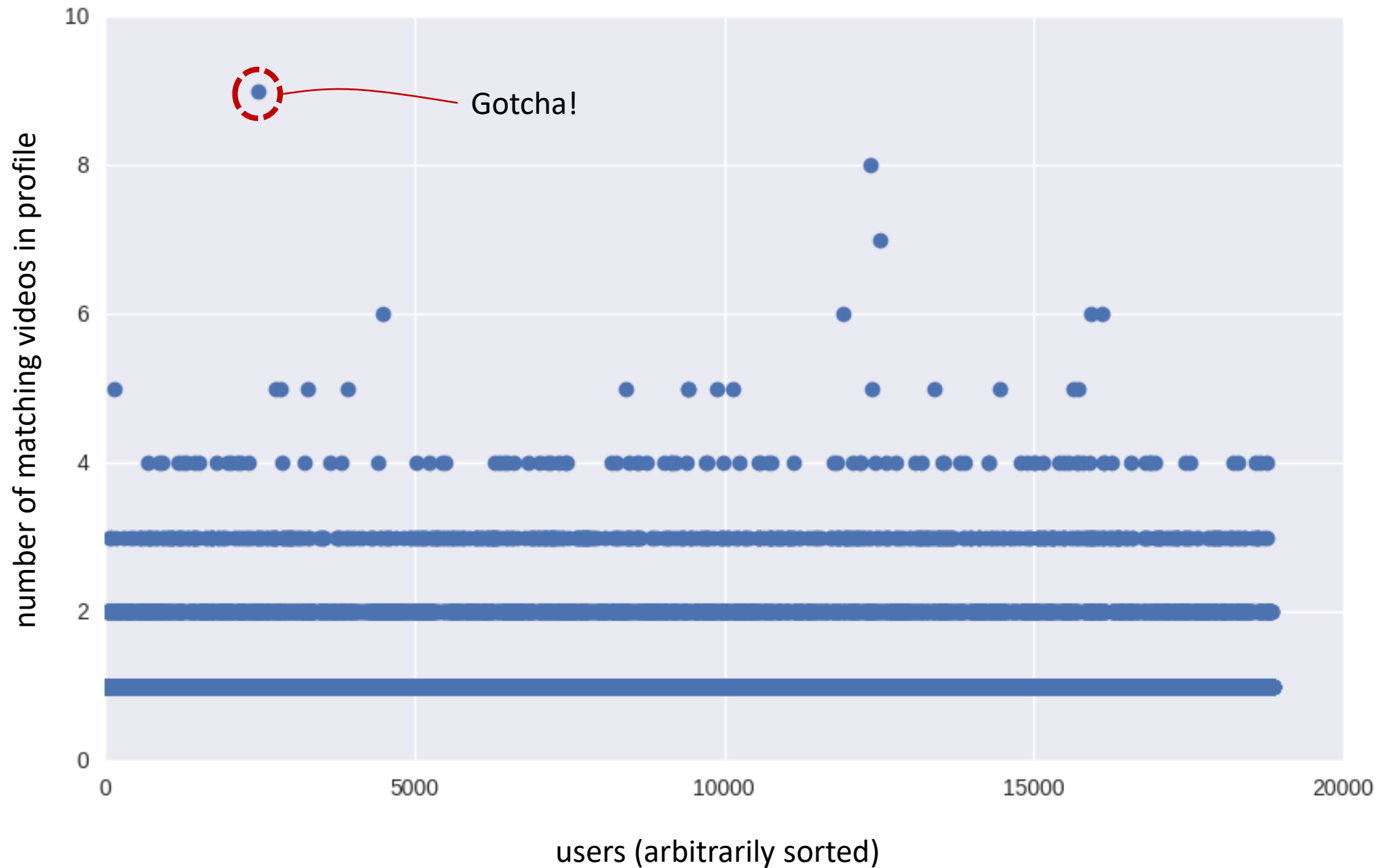
- We download public playlists from users (often linked via Google+)
- We extract the video IDs using the Youtube API
- We feed the resulting video IDs into our algorithm (this time initialized with video IDs instead of domains)



Example

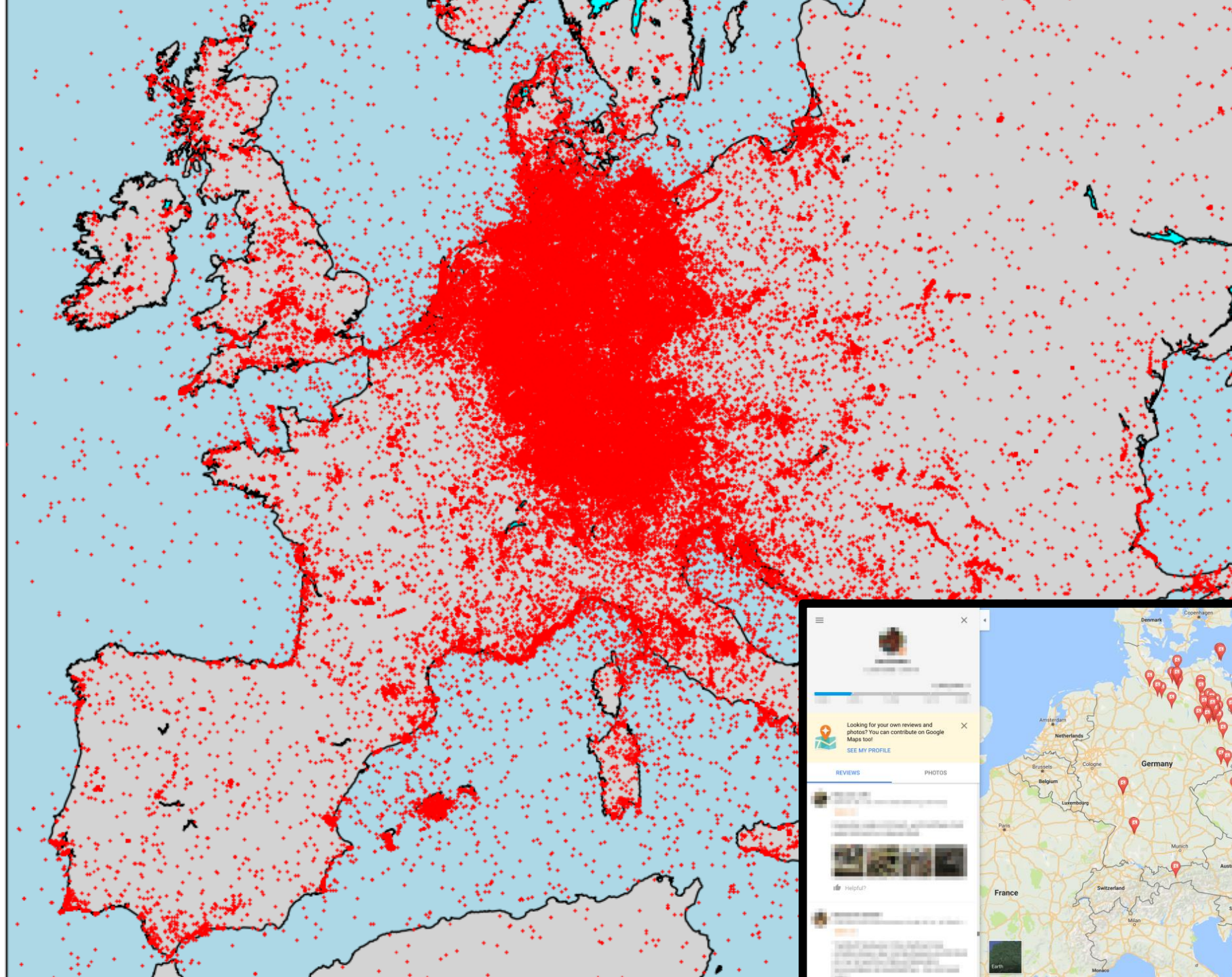
Video-IDs:

02Zm-Ayv-PA
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2ips2mM7Zqw
2wUv1TUi8kQ
34Na4j8AVgA
3VVuMIB2hC0
4fXvJHrbUTA
4ulaGjwiIbo
5BzkbSq7pww
5RDSkR8_AQ0
680R1Gq2YYU
6IHq9yv_qis
8d5QEWdHchk
...



More ways to extract public information

- Google Maps URLs (contain latitude/longitude)
- Facebook Post IDs (URLs were anonymized in the data set but IDs were shared)
- ...



"Instant" deanonymization via a unique URL

The image displays two browser windows side-by-side, illustrating a process of deanonymization. The top window shows a Twitter Analytics profile for 'Mein Account' (Andreas Dewes @japh44). The URL bar highlights the unique identifier 'japh44' in the address 'https://analytics.twitter.com/user/japh44/home'. The bottom window shows a XING profile for 'Dr. Andreas Dewes'. The URL bar highlights a unique identifier 'mx_b_p' in the address 'https://www.xing.com/profile/Andreas_Dewes?sc_o=mx_b_p'. The XING profile also shows a 'PREMIUM' badge and a list of 'Profilbesucher' (profile visitors).

What did we find in the data?

[illegible]

Ladies and Gentlemen, because of an **investigation concerning computer fraud** (file number), which I have dealt with here, § 113 TKG i.V.m. § 100j StPO I need information on following **IP address: xxx.xxx.xxx.xxx Time stamp: xx.xx.2016, 10:05:31 CEST**

The data is needed to identify the offender. Please send your answer by e-mail to the following address

Firstname.lastname@police.state.de or by fax.

first name

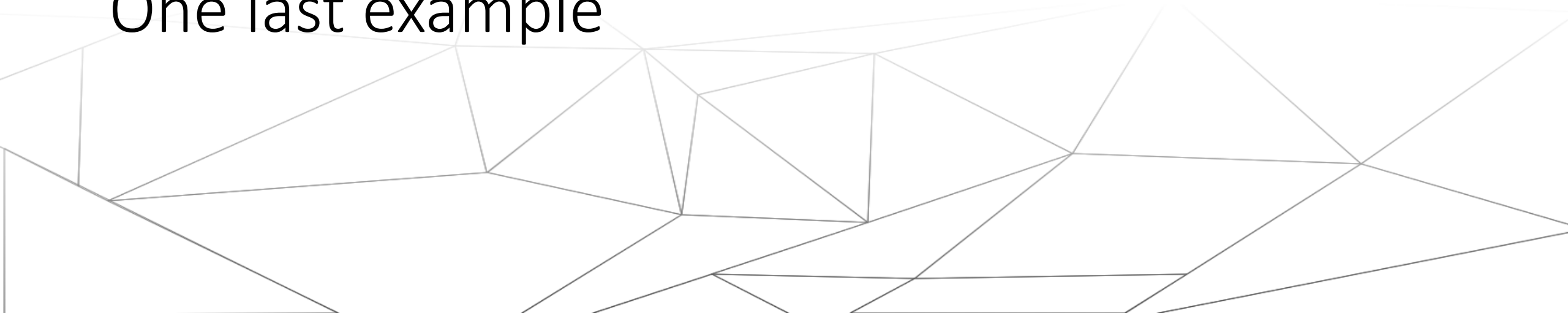
Last Name

Detective Chief Place of **county**

Cybercrime

phone number

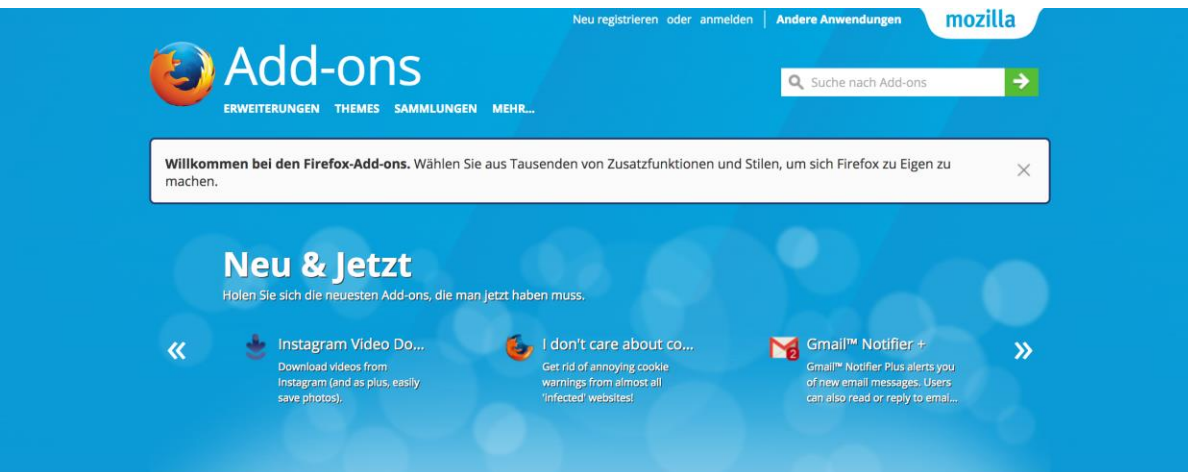
One last example



[illegible]

Who collected the data?

Browser Plugins



Check out our new Mobile App



EN ▾ Login \ Register



NEW!
Mobile

Community

Our APIs

Support

Search a website for its reputation



Web of Trust (MyWOT) Safe Web Search & Browsing

140,000,000



[DATUM] 11:15:04 <http://what.kuketz.de/>

[...]

[DATUM] 15:49:27 <https://www.ebay-kleinanzeigen.de/p-anzeige-bearbeiten.html?adId=xxx>

[DATUM] 13:06:23 <http://what.kuketz.de/>

[...]

[DATUM] 11:22:18 <http://what.kuketz.de/>

[DATUM] 14:59:30 <http://blog.fefe.de/>

[...]

[DATUM] 14:59:36 <http://what.kuketz.de/>

[DATUM] 14:59:44

https://www.mywot.com/en/scorecard/what.kuketz.de?utm_source=addon&utm_content=rw-viewsc

[...]

[DATUM] 13:48:24 <http://what.kuketz.de/>

[...]

test by Mike Kuketz / www.kuketz-blog.de



Protect your Browser

NEW!

Mobile

Community

Our APIs

Support

EN ▼

Login \ Register

Search a website for its reputation



Privacy

Privacy policy

Privacy Policy

3. Automatically Collected Information – When you install and use the WOT app or WOT extension, we also automatically collect information from you related to your use of the WOT Services and your web browsing activity. This information may include, without limitation, the following:

- From WOT desktop/mobile extension and desktop app - Internet Protocol Address; device type; operating system and browser; web pages visited and time stamp of the visit; automatically-generated GUID and WOT user ID.

intended to make our users fully aware of the data we collect, the way it is stored, used, and shared, and our users' rights in relation to such

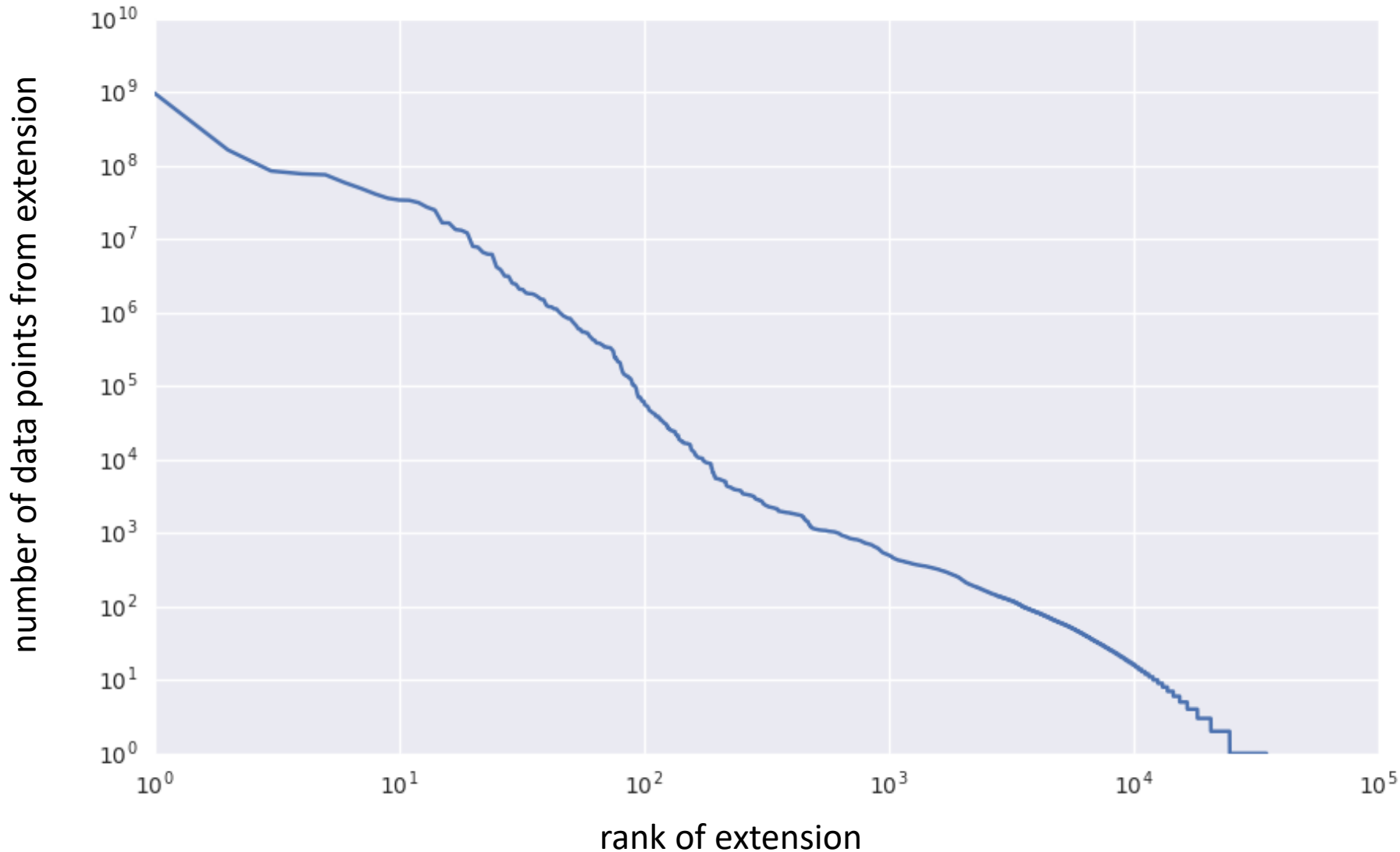
data and practices. We greatly value our users and their privacy rights, and we encourage all of our users to read and become familiar with this

or de-identified information that is not collected together, or otherwise associated by us, with any Registration Information or other Personal Information, if any, that we hold about you. When we automatically collect information, we go to great lengths to make sure that Non-Personal Information remains anonymous, as further described below.

Information and Non-Personal Information in accordance with the terms of this Privacy Policy. **IF YOU DO NOT ACCEPT THE TERM OF THIS PRIVACY POLICY, PLEASE DO NOT ACCESS OR USE THE WOT PRODUCTS OR WOT SERVICES.**

- Information We Collect
- How We Use the Data
- Sharing Collected Data with Third Parties
- Cookies
- Do Not Track Disclosure
- Your Responsibility for User-Generated Content
- Data Storage and Security
- Data Retention

How many extensions are affected?

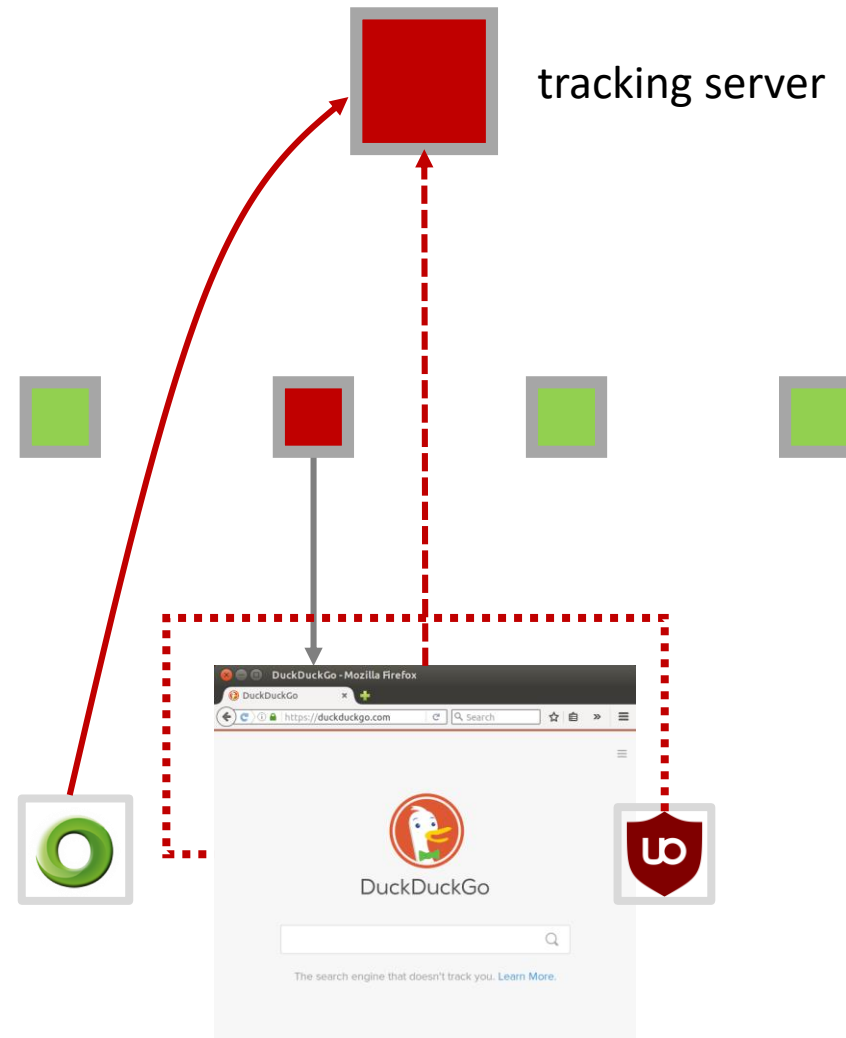


95 % of the data comes from only **10 extension (variants/versions)**.

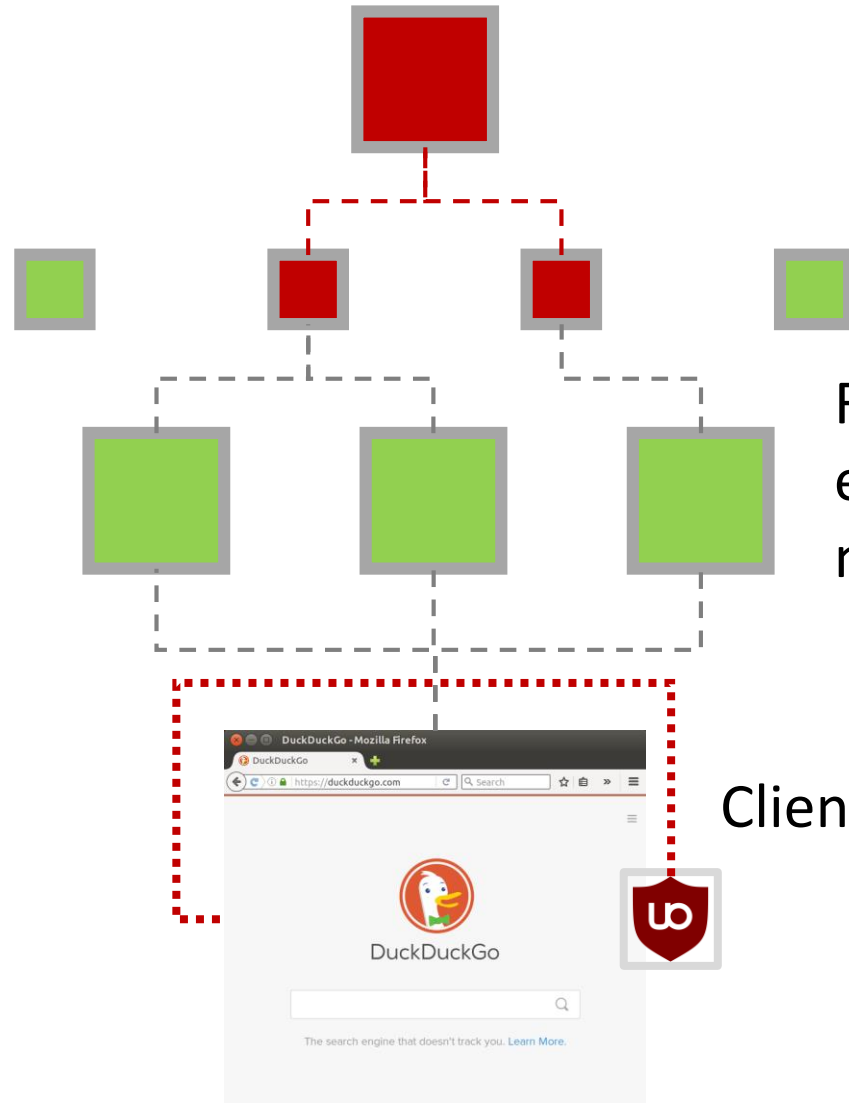
Many more are spying on their users, but have a small installation base.

Up to **10.000 extension versions** affected (upper bound analysis via extension ID).

Why use extensions for tracking?



(How) can I protect myself against tracking?



Rotating proxy servers ($n \gg 1$)
e.g. TOR or a VPN with rotating exit nodes

Client-side blocking of trackers

Can I hide in my data by generating noise?

(e.g. via random page visits)

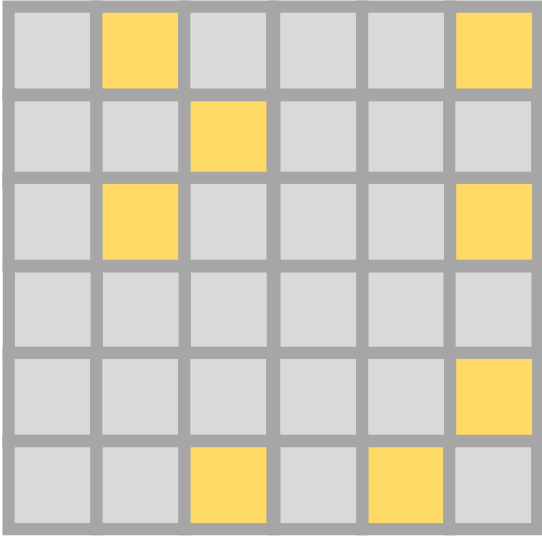
ಠ_ಠ(ツ)ಠ

Usually not

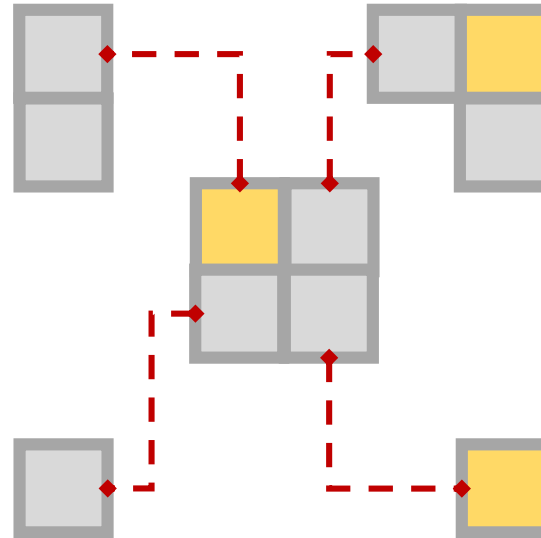


argmax $\|M \cdot v\|$ is robust against isolated (additive) perturbation

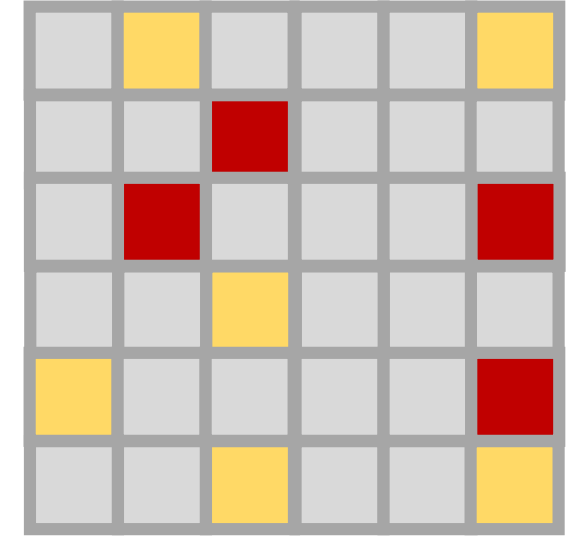
Takeaways



High-dimensional, user-related data is really hard to robustly anonymize (even if you really try to do so).



The increase in publicly available information on many people makes de-anonymization via linkage attacks easier than ever before.



Often, only a few external data points (<10) are sufficient to uniquely identify a person.

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Many further sources and contributors

TV shows ARD Panorama, NDR Panorama3 and ZAPP

<http://daserste.ndr.de/panorama/aktuell/Web-Strip-Intimate-data-from-federal-politicians-for-sale,nacktimnetz114.html>

Questions?

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